基于若依开源平台与通义大模型门店鲜花智能销售推荐系统

1. 系统概述

1.1 项目背景

随着鲜花市场的快速发展和消费者对个性化服务需求的提升，传统的鲜花销售模式已无法满足现代消费者的需求。本项目旨在开发一个基于若依开源平台的门店鲜花智能销售系统，通过智能化管理提升销售效率，优化客户体验。

1.2 系统目标

- 实现鲜花商品的智能化管理
- 提供智能推荐功能，根据客户喜好推荐合适的鲜花
- 支持线上线下一体化销售模式
- 实现销售数据的实时统计与分析
- 提供便捷的订单处理流程
- 支持鲜花预定功能
- 实现完整的销售历史记录追踪

2. 功能需求分析

2.1 核心业务功能

2.1.1 鲜花商品管理

- 鲜花分类管理（如：玫瑰、百合、康乃馨等）
- 鲜花信息维护（名称、描述、价格、库存等）
- 鲜花图片管理
- 鲜花状态管理（上架、下架等）

2.1.2 智能推荐系统

- 基于客户历史购买记录的推荐
- 基于节日、场合的智能推荐
- 基于季节、花语的推荐
- 基于客户偏好的个性化推荐

2.1.3 销售管理

- 订单生成与处理
- 支付管理
- 库存自动扣减
- 销售记录统计
- 销售历史记录查询

2.1.4 预定管理

- 鲜花预定功能
- 预定时间管理
- 预定状态跟踪
- 预定提醒通知

2.1.5 客户管理

- 客户信息维护
- 客户购买历史记录
- 客户偏好设置
- 客户等级管理

2.1.6 库存管理

- 实时库存监控
- 库存预警机制
- 入库出库记录
- 库存调拨管理

2.2 系统非功能需求

- 系统响应时间不超过3秒
- 支持至少1000个并发用户
- 数据备份与恢复机制
- 系统安全性保障

3. 技术架构

3.1 服务端技术栈

- 基于若依开源平台进行数据管理
- Spring Boot + MyBatis 作为后端框架
- MySQL 作为数据库存储

3.2 智能终端技术栈

- MyBatis + Spring Boot 作为后端框架
- LangChain4j 用于智能推荐功能
- Vue + Element Plus 作为前端框架

4. 数据库设计

4.1 数据库表设计原则

- 遵循数据库设计三大范式
- 合理使用索引提高查询效率
- 添加必要的约束保证数据完整性
- 每张表都包含创建时间和更新时间字段
- 所有业务表均以"flower_"作为前缀，与若依平台表区分

4.2 核心业务表设计

4.2.1 鲜花分类表 (flower_category)

存储鲜花的分类信息

4.2.2 鲜花信息表 (flower_info)

存储鲜花的详细信息

4.2.3 客户信息表 (flower_customer)

存储客户的基本信息和偏好

4.2.4 订单表 (flower_order)

存储订单的基本信息

4.2.5 订单明细表 (flower_order_detail)

存储订单中每个商品的详细信息

4.2.6 库存表 (flower_inventory)

存储鲜花的库存信息

4.2.7 推荐记录表 (flower_recommend_record)

存储系统为客户推荐的鲜花记录

4.2.8 预定表 (flower_reservation)

存储鲜花预定信息

4.2.9 销售历史记录表 (flower_sales_history)

存储详细的销售历史记录

4.2.10 客户购买历史表 (flower_customer_purchase_history)

存储客户购买历史记录

5. 数据库表结构与初始化脚本

以下为详细的数据库表结构设计和初始化数据。

5.1 创建数据库

```
CREATE DATABASE IF NOT EXISTS flower_sales_system DEFAULT CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;
USE flower_sales_system;
```

5.2 鲜花分类表 (flower_category)

```
-- 鲜花分类表
DROP TABLE IF EXISTS flower_category;
CREATE TABLE flower_category (
    id BIGINT(20) NOT NULL AUTO_INCREMENT COMMENT '分类ID',
    category_name VARCHAR(50) NOT NULL COMMENT '分类名称',
    category_desc VARCHAR(200) DEFAULT NULL COMMENT '分类描述',
    sort_order INT(11) DEFAULT 0 COMMENT '排序',
```

```
    status TINYINT(1) DEFAULT 1 COMMENT '状态 1-正常 0-禁用',
    create_by VARCHAR(64) DEFAULT '' COMMENT '创建者',
    create_time DATETIME DEFAULT CURRENT_TIMESTAMP COMMENT '创建时间',
    update_by VARCHAR(64) DEFAULT '' COMMENT '更新者',
    update_time DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP COMMENT
'更新时间',
    remark VARCHAR(500) DEFAULT NULL COMMENT '备注',
    PRIMARY KEY (id),
    UNIQUE KEY uk_category_name (category_name)
) ENGINE=InnoDB AUTO_INCREMENT=1 DEFAULT CHARSET=utf8mb4 COMMENT='鲜花分类表';

-- 插入10条模拟数据
INSERT INTO flower_category (category_name, category_desc, sort_order, status, remark)
VALUES
('玫瑰', '玫瑰花系列', 1, 1, '情人节主打'),
('百合', '百合花系列', 2, 1, '婚礼常用'),
('康乃馨', '康乃馨系列', 3, 1, '母亲节主打'),
('郁金香', '郁金香系列', 4, 1, '春季特色'),
('向日葵', '向日葵系列', 5, 1, '阳光主题'),
('满天星', '满天星系列', 6, 1, '配花首选'),
('紫罗兰', '紫罗兰系列', 7, 1, '优雅系列'),
('茉莉花', '茉莉花系列', 8, 1, '清香系列'),
('牡丹', '牡丹系列', 9, 1, '富贵主题'),
('菊花', '菊花系列', 10, 1, '秋季特色');
```

5.3 鲜花信息表 (flower_info)

```
-- 鲜花信息表
DROP TABLE IF EXISTS flower_info;
CREATE TABLE flower_info (
    id BIGINT(20) NOT NULL AUTO_INCREMENT COMMENT '鲜花ID',
    flower_name VARCHAR(100) NOT NULL COMMENT '鲜花名称',
    category_id BIGINT(20) NOT NULL COMMENT '分类ID',
    flower_desc TEXT DEFAULT NULL COMMENT '鲜花描述',
    price DECIMAL(10,2) NOT NULL COMMENT '价格',
    market_price DECIMAL(10,2) DEFAULT NULL COMMENT '市场价',
    cost_price DECIMAL(10,2) DEFAULT NULL COMMENT '成本价',
    image_urls TEXT DEFAULT NULL COMMENT '图片URL列表(JSON格式)',
    qr_code_url VARCHAR(255) DEFAULT NULL COMMENT '二维码URL',
    stock INT(11) DEFAULT 0 COMMENT '库存数量',
    sales_count INT(11) DEFAULT 0 COMMENT '销售数量',
    status TINYINT(1) DEFAULT 1 COMMENT '状态 1-上架 0-下架',
    create_by VARCHAR(64) DEFAULT '' COMMENT '创建者',
    create_time DATETIME DEFAULT CURRENT_TIMESTAMP COMMENT '创建时间',
    update_by VARCHAR(64) DEFAULT '' COMMENT '更新者',
    update_time DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP COMMENT
'更新时间',
    remark VARCHAR(500) DEFAULT NULL COMMENT '备注',
    PRIMARY KEY (id),
    KEY idx_category_id (category_id),
    KEY idx_status (status),
    CONSTRAINT fk_flower_category FOREIGN KEY (category_id) REFERENCES
flower_category(id)
) ENGINE=InnoDB AUTO_INCREMENT=1 DEFAULT CHARSET=utf8mb4 COMMENT='鲜花信息表';

-- 插入10条模拟数据（使用百度网络图片）
INSERT INTO flower_info (flower_name, category_id, flower_desc, price, market_price,
cost_price, image_urls, qr_code_url, stock, sales_count, status, remark) VALUES
```

```
('红玫瑰', 1, '红色玫瑰，象征热情的爱情', 99.00, 120.00, 50.00,
'[\"https://bkimg.cdn.bcebos.com/pic/0823dd54564e9258d10969f69c82d158ccbf4e02?x-bce-process=image/resize,m_lfit,w_536,limit_1/format,f_auto\"]',
'https://example.com/qrcode/rose_red', 100, 50, 1, '经典款'),
('白百合', 2, '白色百合，象征纯洁和高雅', 88.00, 110.00, 45.00,
'[\"https://bkimg.cdn.bcebos.com/pic/d058ccbf6c81800a19d554b4b03533fa828b4702?x-bce-process=image/resize,m_lfit,w_536,limit_1/format,f_auto\"]',
'https://example.com/qrcode/lily_white', 80, 30, 1, '婚礼首选'),
('粉康乃馨', 3, '粉色康乃馨，表达温馨的祝福', 66.00, 80.00, 35.00,
'[\"https://bkimg.cdn.bcebos.com/pic/09fa513d269759ee3d6ac93db4fb43166d22df16?x-bce-process=image/resize,m_lfit,w_536,limit_1/format,f_auto\"]',
'https://example.com/qrcode/carnation_pink', 120, 70, 1, '母亲节特供'),
('黄郁金香', 4, '黄色郁金香，象征阳光和快乐', 77.00, 95.00, 40.00,
'[\"https://bkimg.cdn.bcebos.com/pic/500fd9f9d72a6059252f7a9e2734349b033bba16?x-bce-process=image/resize,m_lfit,w_536,limit_1/format,f_auto\"]',
'https://example.com/qrcode/tulip_yellow', 90, 40, 1, '春季限定'),
('橙色向日葵', 5, '橙色向日葵，传递积极向上的能量', 55.00, 70.00, 30.00,
'[\"https://bkimg.cdn.bcebos.com/pic/35a85edf8db1cb134954e1dcd254564e92584b02?x-bce-process=image/resize,m_lfit,w_536,limit_1/format,f_auto\"]',
'https://example.com/qrcode/sunflower_orange', 60, 25, 1, '活力系列'),
('紫色满天星', 6, '紫色满天星，浪漫的配花选择', 35.00, 45.00, 20.00,
'[\"https://bkimg.cdn.bcebos.com/pic/960a304e251f95ca7b3e6e75ce177f3e67095202?x-bce-process=image/resize,m_lfit,w_536,limit_1/format,f_auto\"]',
'https://example.com/qrcode/babybreath_purple', 200, 100, 1, '配花必备'),
('蓝紫罗兰', 7, '蓝紫罗兰，代表忠诚和谦逊', 68.00, 85.00, 38.00,
'[\"https://bkimg.cdn.bcebos.com/pic/4034970a304e251f95ca7b3e6e75ce177f3e6709?x-bce-process=image/resize,m_lfit,w_536,limit_1/format,f_auto\"]',
'https://example.com/qrcode/violet_blue', 70, 35, 1, '优雅之选'),
('白茉莉', 8, '白色茉莉，散发着淡雅的清香', 45.00, 60.00, 25.00,
'[\"https://bkimg.cdn.bcebos.com/pic/0df41bd5ad6eddc451da692d41dbb6fd52663302?x-bce-process=image/resize,m_lfit,w_536,limit_1/format,f_auto\"]',
'https://example.com/qrcode/jasmine_white', 110, 55, 1, '清香系列'),
('红牡丹', 9, '红色牡丹，富贵吉祥的象征', 120.00, 150.00, 70.00,
'[\"https://bkimg.cdn.bcebos.com/pic/d62a6059252dd42a28349b34003b5bb5c9eab802?x-bce-process=image/resize,m_lfit,w_536,limit_1/format,f_auto\"]',
'https://example.com/qrcode/peony_red', 50, 20, 1, '富贵花开'),
('黄菊花', 10, '黄色菊花，秋季盛开的佳品', 40.00, 55.00, 22.00,
'[\"https://bkimg.cdn.bcebos.com/pic/0824ab18972bd40735ad8a1d7b899e510fb30916?x-bce-process=image/resize,m_lfit,w_536,limit_1/format,f_auto\"]',
'https://example.com/qrcode/chrysanthemum_yellow', 150, 80, 1, '秋季特供');
```

5.4 客户信息表 (flower_customer)

```sql
-- 客户信息表
DROP TABLE IF EXISTS flower_customer;
CREATE TABLE flower_customer (
    id BIGINT(20) NOT NULL AUTO_INCREMENT COMMENT '客户ID',
    customer_name VARCHAR(50) NOT NULL COMMENT '客户姓名',
    phone VARCHAR(20) DEFAULT NULL COMMENT '联系电话',
    email VARCHAR(100) DEFAULT NULL COMMENT '邮箱',
    address VARCHAR(200) DEFAULT NULL COMMENT '地址',
    birthday DATE DEFAULT NULL COMMENT '生日',
    gender TINYINT(1) DEFAULT 1 COMMENT '性别 1-男 0-女',
    preference TEXT DEFAULT NULL COMMENT '客户偏好（JSON格式存储）',
    level TINYINT(1) DEFAULT 1 COMMENT '客户等级 1-普通 2-银卡 3-金卡 4-钻石',
    points INT(11) DEFAULT 0 COMMENT '积分',
    status TINYINT(1) DEFAULT 1 COMMENT '状态 1-正常 0-禁用',
    create_by VARCHAR(64) DEFAULT '' COMMENT '创建者',
    create_time DATETIME DEFAULT CURRENT_TIMESTAMP COMMENT '创建时间',
```

```
    update_by VARCHAR(64) DEFAULT '' COMMENT '更新者',
    update_time DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP COMMENT
'更新时间',
    remark VARCHAR(500) DEFAULT NULL COMMENT '备注',
    PRIMARY KEY (id),
    UNIQUE KEY uk_phone (phone),
    KEY idx_customer_name (customer_name)
) ENGINE=InnoDB AUTO_INCREMENT=1 DEFAULT CHARSET=utf8mb4 COMMENT='客户信息表';

-- 插入10条模拟数据
INSERT INTO flower_customer (customer_name, phone, email, address, birthday, gender,
preference, level, points, status, remark) VALUES
('张三', '13800138001', 'zhangsan@example.com', '北京市朝阳区xxx街道', '1990-05-15', 1,
'{"favorite_color":"红色","preferred_occasion":"情人节","preferred_category":[1,2]}', 3,
1200, 1, '金卡客户'),
('李四', '13800138002', 'lisi@example.com', '上海市浦东新区xxx路', '1985-12-10', 0,
'{"favorite_color":"粉色","preferred_occasion":"母亲节","preferred_category":[3,6]}', 2,
800, 1, '银卡客户'),
('王五', '13800138003', 'wangwu@example.com', '广州市天河区xxx大道', '1992-03-22', 1,
'{"favorite_color":"白色","preferred_occasion":"婚礼","preferred_category":[2,9]}', 4,
2500, 1, '钻石客户'),
('赵六', '13800138004', 'zhaoliu@example.com', '深圳市南山区xxx路', '1988-07-30', 0,
'{"favorite_color":"紫色","preferred_occasion":"生日","preferred_category":[5,7]}', 1,
200, 1, '普通客户'),
('钱七', '13800138005', 'qianqi@example.com', '杭州市西湖区xxx街道', '1995-11-11', 1,
'{"favorite_color":"黄色","preferred_occasion":"开业","preferred_category":[4,5]}', 2,
600, 1, '银卡客户'),
('孙八', '13800138006', 'sunba@example.com', '南京市鼓楼区xxx路', '1991-09-09', 0,
'{"favorite_color":"蓝色","preferred_occasion":"纪念日","preferred_category":[7,8]}', 3,
1500, 1, '金卡客户'),
('周九', '13800138007', 'zhoujiu@example.com', '成都市锦江区xxx街道', '1987-04-18', 1,
'{"favorite_color":"橙色","preferred_occasion":"探病","preferred_category":[3,10]}', 1,
100, 1, '普通客户'),
('吴十', '13800138008', 'wushi@example.com', '武汉市江汉区xxx大道', '1993-06-25', 0,
'{"favorite_color":"绿色","preferred_occasion":"毕业","preferred_category":[1,8]}', 2,
700, 1, '银卡客户'),
('郑十一', '13800138009', 'zhengshiyi@example.com', '西安市雁塔区xxx路', '1994-02-14', 1,
'{"favorite_color":"粉色","preferred_occasion":"表白","preferred_category":[1,6]}', 3,
1300, 1, '金卡客户'),
('王十二', '13800138010', 'wangshier@example.com', '重庆市渝中区xxx街道', '1989-08-08', 0,
'{"favorite_color":"红色","preferred_occasion":"结婚","preferred_category":[1,2]}', 4,
3000, 1, '钻石客户');
```

5.5 订单表 (flower_order)

```
-- 订单表
DROP TABLE IF EXISTS flower_order;
CREATE TABLE flower_order (
    id BIGINT(20) NOT NULL AUTO_INCREMENT COMMENT '订单ID',
    order_no VARCHAR(32) NOT NULL COMMENT '订单编号',
    customer_id BIGINT(20) NOT NULL COMMENT '客户ID',
    order_type TINYINT(1) DEFAULT 1 COMMENT '订单类型 1-普通订单 2-预定订单',
    order_amount DECIMAL(10,2) NOT NULL COMMENT '订单金额',
    discount_amount DECIMAL(10,2) DEFAULT 0.00 COMMENT '优惠金额',
    actual_amount DECIMAL(10,2) NOT NULL COMMENT '实际支付金额',
    order_status TINYINT(1) DEFAULT 1 COMMENT '订单状态 1-待付款 2-已付款 3-已发货 4-已完成
5-已取消',
    payment_method TINYINT(1) DEFAULT 1 COMMENT '支付方式 1-微信 2-支付宝 3-银行卡 4-现金',
    receiver_name VARCHAR(50) DEFAULT NULL COMMENT '收货人姓名',
```

```
    receiver_phone VARCHAR(20) DEFAULT NULL COMMENT '收货人电话',
    receiver_address VARCHAR(200) DEFAULT NULL COMMENT '收货地址',
    delivery_time DATETIME DEFAULT NULL COMMENT '发货时间',
    receive_time DATETIME DEFAULT NULL COMMENT '收货时间',
    qr_code_url VARCHAR(255) DEFAULT NULL COMMENT '订单二维码URL',
    create_by VARCHAR(64) DEFAULT '' COMMENT '创建者',
    create_time DATETIME DEFAULT CURRENT_TIMESTAMP COMMENT '创建时间',
    update_by VARCHAR(64) DEFAULT '' COMMENT '更新者',
    update_time DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP COMMENT
'更新时间',
    remark VARCHAR(500) DEFAULT NULL COMMENT '备注',
    PRIMARY KEY (id),
    UNIQUE KEY uk_order_no (order_no),
    KEY idx_customer_id (customer_id),
    KEY idx_order_status (order_status),
    CONSTRAINT fk_order_customer FOREIGN KEY (customer_id) REFERENCES
flower_customer(id)
) ENGINE=InnoDB AUTO_INCREMENT=1 DEFAULT CHARSET=utf8mb4 COMMENT='订单表';

-- 插入10条模拟数据
INSERT INTO flower_order (order_no, customer_id, order_type, order_amount,
discount_amount, actual_amount, order_status, payment_method, receiver_name,
receiver_phone, receiver_address, delivery_time, receive_time, qr_code_url, remark)
VALUES
('ORD202309010001', 1, 1, 198.00, 20.00, 178.00, 4, 1, '张三', '13800138001', '北京市朝阳
区xxx街道', '2023-09-01 10:30:00', '2023-09-02 14:20:00',
'https://example.com/qrcode/order/ORD202309010001', '情人节订单'),
('ORD202309020002', 2, 1, 88.00, 0.00, 88.00, 4, 2, '李四', '13800138002', '上海市浦东新区
xxx路', '2023-09-02 09:15:00', '2023-09-03 11:45:00',
'https://example.com/qrcode/order/ORD202309020002', '母亲节订单'),
('ORD202309030003', 3, 1, 240.00, 50.00, 190.00, 4, 1, '王五', '13800138003', '广州市天河
区xxx大道', '2023-09-03 14:20:00', '2023-09-04 16:30:00',
'https://example.com/qrcode/order/ORD202309030003', '金卡客户订单'),
('ORD202309040004', 4, 1, 55.00, 0.00, 55.00, 4, 3, '赵六', '13800138004', '深圳市南山区
xxx路', '2023-09-04 11:10:00', '2023-09-05 13:40:00',
'https://example.com/qrcode/order/ORD202309040004', '开业花篮'),
('ORD202309050005', 5, 1, 134.00, 10.00, 124.00, 4, 2, '钱七', '13800138005', '杭州市西湖
区xxx街道', '2023-09-05 15:45:00', '2023-09-06 10:20:00',
'https://example.com/qrcode/order/ORD202309050005', '生日订单'),
('ORD202309060006', 6, 2, 113.00, 0.00, 113.00, 3, 1, '孙八', '13800138006', '南京市鼓楼区
xxx路', '2023-09-06 13:30:00', NULL, 'https://example.com/qrcode/order/ORD202309060006',
'纪念日预定订单'),
('ORD202309070007', 7, 1, 40.00, 5.00, 35.00, 2, 4, '周九', '13800138007', '成都市锦江区
xxx街道', NULL, NULL, 'https://example.com/qrcode/order/ORD202309070007', '探病订单'),
('ORD202309080008', 8, 1, 155.00, 0.00, 155.00, 1, 2, '吴十', '13800138008', '武汉市江汉区
xxx大道', NULL, NULL, 'https://example.com/qrcode/order/ORD202309080008', '毕业季订单'),
('ORD202309090009', 9, 2, 218.00, 30.00, 188.00, 2, 1, '郑十一', '13800138009', '西安市雁
塔区xxx路', NULL, NULL, 'https://example.com/qrcode/order/ORD202309090009', '表白预定订
单'),
('ORD202309100010', 10, 1, 300.00, 100.00, 200.00, 4, 2, '王十二', '13800138010', '重庆市
渝中区xxx街道', '2023-09-10 16:20:00', '2023-09-11 09:45:00',
'https://example.com/qrcode/order/ORD202309100010', '钻石客户订单');
```

5.6 订单明细表 (flower_order_detail)

```
-- 订单明细表
DROP TABLE IF EXISTS flower_order_detail;
CREATE TABLE flower_order_detail (
    id BIGINT(20) NOT NULL AUTO_INCREMENT COMMENT '明细ID',
```

```
    order_id BIGINT(20) NOT NULL COMMENT '订单ID',
    flower_id BIGINT(20) NOT NULL COMMENT '鲜花ID',
    quantity INT(11) NOT NULL COMMENT '购买数量',
    unit_price DECIMAL(10,2) NOT NULL COMMENT '单价',
    total_price DECIMAL(10,2) NOT NULL COMMENT '总价',
    create_by VARCHAR(64) DEFAULT '' COMMENT '创建者',
    create_time DATETIME DEFAULT CURRENT_TIMESTAMP COMMENT '创建时间',
    update_by VARCHAR(64) DEFAULT '' COMMENT '更新者',
    update_time DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP COMMENT
'更新时间',
    remark VARCHAR(500) DEFAULT NULL COMMENT '备注',
    PRIMARY KEY (id),
    KEY idx_order_id (order_id),
    KEY idx_flower_id (flower_id),
    CONSTRAINT fk_detail_order FOREIGN KEY (order_id) REFERENCES flower_order(id),
    CONSTRAINT fk_detail_flower FOREIGN KEY (flower_id) REFERENCES flower_info(id)
) ENGINE=InnoDB AUTO_INCREMENT=1 DEFAULT CHARSET=utf8mb4 COMMENT='订单明细表';

-- 插入模拟数据（每个订单2个明细项）
INSERT INTO flower_order_detail (order_id, flower_id, quantity, unit_price, total_price,
remark) VALUES
(1, 1, 2, 99.00, 198.00, '红玫瑰*2'),
(1, 6, 1, 35.00, 35.00, '满天星配花'),
(2, 2, 1, 88.00, 88.00, '白百合'),
(3, 9, 2, 120.00, 240.00, '红牡丹*2'),
(4, 5, 1, 55.00, 55.00, '向日葵'),
(5, 1, 1, 99.00, 99.00, '红玫瑰'),
(5, 4, 1, 77.00, 77.00, '郁金香'),
(6, 3, 1, 66.00, 66.00, '康乃馨'),
(6, 7, 1, 68.00, 68.00, '紫罗兰'),
(7, 10, 1, 40.00, 40.00, '菊花'),
(8, 8, 1, 45.00, 45.00, '茉莉花'),
(8, 3, 2, 66.00, 132.00, '康乃馨*2'),
(9, 1, 2, 99.00, 198.00, '红玫瑰*2'),
(10, 9, 1, 120.00, 120.00, '红牡丹'),
(10, 1, 2, 99.00, 198.00, '红玫瑰*2'),
(10, 6, 3, 35.00, 105.00, '满天星*3');
```

5.7 库存表 (flower_inventory)

```
-- 库存表
DROP TABLE IF EXISTS flower_inventory;
CREATE TABLE flower_inventory (
    id BIGINT(20) NOT NULL AUTO_INCREMENT COMMENT '库存ID',
    flower_id BIGINT(20) NOT NULL COMMENT '鲜花ID',
    quantity INT(11) NOT NULL COMMENT '库存数量',
    reserved_quantity INT(11) DEFAULT 0 COMMENT '预定占用数量',
    available_quantity INT(11) NOT NULL COMMENT '可用数量',
    warning_threshold INT(11) DEFAULT 10 COMMENT '预警阈值',
    last_restock_time DATETIME DEFAULT NULL COMMENT '最近补货时间',
    create_by VARCHAR(64) DEFAULT '' COMMENT '创建者',
    create_time DATETIME DEFAULT CURRENT_TIMESTAMP COMMENT '创建时间',
    update_by VARCHAR(64) DEFAULT '' COMMENT '更新者',
    update_time DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP COMMENT
'更新时间',
    remark VARCHAR(500) DEFAULT NULL COMMENT '备注',
    PRIMARY KEY (id),
    UNIQUE KEY uk_flower_id (flower_id),
    CONSTRAINT fk_inventory_flower FOREIGN KEY (flower_id) REFERENCES flower_info(id)
```

```
) ENGINE=InnoDB AUTO_INCREMENT=1 DEFAULT CHARSET=utf8mb4 COMMENT='库存表';

-- 插入10条模拟数据
INSERT INTO flower_inventory (flower_id, quantity, reserved_quantity,
available_quantity, warning_threshold, last_restock_time, remark) VALUES
(1, 100, 5, 95, 10, '2023-08-25 10:00:00', '红玫瑰库存'),
(2, 80, 2, 78, 10, '2023-08-26 11:00:00', '白百合库存'),
(3, 120, 3, 117, 15, '2023-08-27 09:30:00', '康乃馨库存'),
(4, 90, 1, 89, 10, '2023-08-28 14:20:00', '郁金香库存'),
(5, 60, 2, 58, 10, '2023-08-29 16:45:00', '向日葵库存'),
(6, 200, 10, 190, 20, '2023-08-30 10:15:00', '满天星库存'),
(7, 70, 1, 69, 10, '2023-08-31 11:30:00', '紫罗兰库存'),
(8, 110, 1, 109, 15, '2023-09-01 13:45:00', '茉莉花库存'),
(9, 50, 3, 47, 5, '2023-09-02 15:20:00', '牡丹库存'),
(10, 150, 2, 148, 20, '2023-09-03 09:50:00', '菊花库存');
```

5.8 推荐记录表 (flower_recommend_record)

```
-- 推荐记录表
DROP TABLE IF EXISTS flower_recommend_record;
CREATE TABLE flower_recommend_record (
    id BIGINT(20) NOT NULL AUTO_INCREMENT COMMENT '推荐记录ID',
    customer_id BIGINT(20) NOT NULL COMMENT '客户ID',
    flower_id BIGINT(20) NOT NULL COMMENT '鲜花ID',
    recommend_reason VARCHAR(200) DEFAULT NULL COMMENT '推荐理由',
    recommend_type TINYINT(1) DEFAULT 1 COMMENT '推荐类型 1-节日推荐 2-偏好推荐 3-热销推荐 4-
新品推荐 5-搭配推荐',
    is_accepted TINYINT(1) DEFAULT 0 COMMENT '是否接受 1-接受 0-拒绝',
    acceptance_time DATETIME DEFAULT NULL COMMENT '接受时间',
    create_by VARCHAR(64) DEFAULT '' COMMENT '创建者',
    create_time DATETIME DEFAULT CURRENT_TIMESTAMP COMMENT '创建时间',
    update_by VARCHAR(64) DEFAULT '' COMMENT '更新者',
    update_time DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP COMMENT
'更新时间',
    remark VARCHAR(500) DEFAULT NULL COMMENT '备注',
    PRIMARY KEY (id),
    KEY idx_customer_id (customer_id),
    KEY idx_flower_id (flower_id),
    CONSTRAINT fk_recommend_customer FOREIGN KEY (customer_id) REFERENCES
flower_customer(id),
    CONSTRAINT fk_recommend_flower FOREIGN KEY (flower_id) REFERENCES flower_info(id)
) ENGINE=InnoDB AUTO_INCREMENT=1 DEFAULT CHARSET=utf8mb4 COMMENT='推荐记录表';

-- 插入10条模拟数据
INSERT INTO flower_recommend_record (customer_id, flower_id, recommend_reason,
recommend_type, is_accepted, acceptance_time, remark) VALUES
(1, 1, '情人节推荐', 1, 1, '2023-09-01 09:30:00', '客户接受了推荐'),
(2, 3, '母亲节推荐', 1, 1, '2023-09-02 08:45:00', '客户购买了推荐商品'),
(3, 9, '金卡客户专享推荐', 3, 1, '2023-09-03 10:15:00', '高价值客户推荐'),
(4, 5, '开业场合推荐', 2, 1, '2023-09-04 09:20:00', '根据客户偏好推荐'),
(5, 1, '生日推荐', 1, 0, NULL, '客户未接受推荐'),
(6, 7, '纪念日推荐', 1, 1, '2023-09-06 11:30:00', '客户接受了推荐'),
(7, 10, '探病场合推荐', 2, 1, '2023-09-07 14:45:00', '根据场合推荐'),
(8, 8, '毕业季推荐', 1, 0, NULL, '客户未接受推荐'),
(9, 1, '表白场景推荐', 1, 1, '2023-09-09 08:30:00', '客户接受了推荐'),
(10, 9, '钻石客户专享推荐', 3, 1, '2023-09-10 15:20:00', '高价值客户推荐');
```

5.9 预定表 (flower_reservation)

```sql
-- 预定表
DROP TABLE IF EXISTS flower_reservation;
CREATE TABLE flower_reservation (
    id BIGINT(20) NOT NULL AUTO_INCREMENT COMMENT '预定ID',
    reservation_no VARCHAR(32) NOT NULL COMMENT '预定编号',
    customer_id BIGINT(20) NOT NULL COMMENT '客户ID',
    flower_id BIGINT(20) NOT NULL COMMENT '鲜花ID',
    quantity INT(11) NOT NULL COMMENT '预定数量',
    unit_price DECIMAL(10,2) NOT NULL COMMENT '单价',
    total_amount DECIMAL(10,2) NOT NULL COMMENT '总金额',
    reservation_date DATE NOT NULL COMMENT '预定日期',
    delivery_time DATETIME DEFAULT NULL COMMENT '配送时间',
    reservation_status TINYINT(1) DEFAULT 1 COMMENT '预定状态 1-待确认 2-已确认 3-已完成 4-已取消',
    contact_person VARCHAR(50) DEFAULT NULL COMMENT '联系人',
    contact_phone VARCHAR(20) DEFAULT NULL COMMENT '联系电话',
    delivery_address VARCHAR(200) DEFAULT NULL COMMENT '配送地址',
    qr_code_url VARCHAR(255) DEFAULT NULL COMMENT '预定二维码URL',
    create_by VARCHAR(64) DEFAULT '' COMMENT '创建者',
    create_time DATETIME DEFAULT CURRENT_TIMESTAMP COMMENT '创建时间',
    update_by VARCHAR(64) DEFAULT '' COMMENT '更新者',
    update_time DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP COMMENT '更新时间',
    remark VARCHAR(500) DEFAULT NULL COMMENT '备注',
    PRIMARY KEY (id),
    UNIQUE KEY uk_reservation_no (reservation_no),
    KEY idx_customer_id (customer_id),
    KEY idx_flower_id (flower_id),
    KEY idx_reservation_status (reservation_status),
    CONSTRAINT fk_reservation_customer FOREIGN KEY (customer_id) REFERENCES flower_customer(id),
    CONSTRAINT fk_reservation_flower FOREIGN KEY (flower_id) REFERENCES flower_info(id)
) ENGINE=InnoDB AUTO_INCREMENT=1 DEFAULT CHARSET=utf8mb4 COMMENT='预定表';

-- 插入10条模拟数据
INSERT INTO flower_reservation (reservation_no, customer_id, flower_id, quantity, unit_price, total_amount, reservation_date, delivery_time, reservation_status, contact_person, contact_phone, delivery_address, qr_code_url, remark) VALUES
('RES202309060001', 6, 3, 1, 66.00, 66.00, '2023-09-20', '2023-09-20 09:00:00', 2, '孙八', '13800138006', '南京市鼓楼区xxx路', 'https://example.com/qrcode/reservation/RES202309060001', '纪念日预定'),
('RES202309090002', 9, 1, 2, 99.00, 198.00, '2023-09-25', '2023-09-25 18:00:00', 2, '郑十一', '13800138009', '西安市雁塔区xxx路', 'https://example.com/qrcode/reservation/RES202309090002', '表白预定'),
('RES202309100003', 1, 2, 1, 88.00, 88.00, '2023-09-30', '2023-09-30 10:00:00', 1, '张三', '13800138001', '北京市朝阳区xxx街道', 'https://example.com/qrcode/reservation/RES202309100003', '国庆节预定'),
('RES202309110004', 3, 9, 1, 120.00, 120.00, '2023-10-01', '2023-10-01 09:00:00', 1, '王五', '13800138003', '广州市天河区xxx大道', 'https://example.com/qrcode/reservation/RES202309110004', '国庆节预定'),
('RES202309120005', 5, 4, 1, 55.00, 55.00, '2023-09-28', '2023-09-28 14:00:00', 2, '钱七', '13800138005', '杭州市西湖区xxx街道', 'https://example.com/qrcode/reservation/RES202309120005', '乔迁预定'),
('RES202309130006', 2, 6, 2, 35.00, 70.00, '2023-09-22', '2023-09-22 15:00:00', 3, '李四', '13800138002', '上海市浦东新区xxx路', '已完成预定'),
```

```
('RES202309140007', 8, 8, 1, 45.00, 45.00, '2023-09-26', '2023-09-26 11:00:00', 1, '吴十', '13800138008', '武汉市江汉区xxx大道',
'https://example.com/qrcode/reservation/RES202309140007', '教师节预定'),
('RES202309150008', 4, 5, 1, 55.00, 55.00, '2023-09-29', '2023-09-29 16:00:00', 2, '赵六', '13800138004', '深圳市南山区xxx路',
'https://example.com/qrcode/reservation/RES202309150008', '中秋节预定'),
('RES202309160009', 10, 1, 3, 99.00, 297.00, '2023-10-05', '2023-10-05 10:00:00', 1, '王十二', '13800138010', '重庆市渝中区xxx街道',
'https://example.com/qrcode/reservation/RES202309160009', '重阳节预定'),
('RES202309170010', 7, 10, 1, 40.00, 40.00, '2023-09-23', '2023-09-23 13:00:00', 4, '周九', '13800138007', '成都市锦江区xxx街道',
'https://example.com/qrcode/reservation/RES202309170010', '已取消预定');
```

5.10 销售历史记录表 (flower_sales_history)

```
-- 销售历史记录表
DROP TABLE IF EXISTS flower_sales_history;
CREATE TABLE flower_sales_history (
    id BIGINT(20) NOT NULL AUTO_INCREMENT COMMENT '销售历史ID',
    order_id BIGINT(20) NOT NULL COMMENT '订单ID',
    flower_id BIGINT(20) NOT NULL COMMENT '鲜花ID',
    customer_id BIGINT(20) NOT NULL COMMENT '客户ID',
    quantity INT(11) NOT NULL COMMENT '销售数量',
    unit_price DECIMAL(10,2) NOT NULL COMMENT '单价',
    total_amount DECIMAL(10,2) NOT NULL COMMENT '总金额',
    sale_date DATE NOT NULL COMMENT '销售日期',
    sale_time DATETIME NOT NULL COMMENT '销售时间',
    payment_method TINYINT(1) DEFAULT 1 COMMENT '支付方式 1-微信 2-支付宝 3-银行卡 4-现金',
    salesperson VARCHAR(50) DEFAULT NULL COMMENT '销售人员',
    create_by VARCHAR(64) DEFAULT '' COMMENT '创建者',
    create_time DATETIME DEFAULT CURRENT_TIMESTAMP COMMENT '创建时间',
    update_by VARCHAR(64) DEFAULT '' COMMENT '更新者',
    update_time DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP COMMENT
'更新时间',
    remark VARCHAR(500) DEFAULT NULL COMMENT '备注',
    PRIMARY KEY (id),
    KEY idx_order_id (order_id),
    KEY idx_flower_id (flower_id),
    KEY idx_customer_id (customer_id),
    KEY idx_sale_date (sale_date),
    CONSTRAINT fk_sales_order FOREIGN KEY (order_id) REFERENCES flower_order(id),
    CONSTRAINT fk_sales_flower FOREIGN KEY (flower_id) REFERENCES flower_info(id),
    CONSTRAINT fk_sales_customer FOREIGN KEY (customer_id) REFERENCES
flower_customer(id)
) ENGINE=InnoDB AUTO_INCREMENT=1 DEFAULT CHARSET=utf8mb4 COMMENT='销售历史记录表';

-- 插入10条模拟数据
INSERT INTO flower_sales_history (order_id, flower_id, customer_id, quantity,
unit_price, total_amount, sale_date, sale_time, payment_method, salesperson, remark)
VALUES
(1, 1, 1, 2, 99.00, 198.00, '2023-09-01', '2023-09-01 10:30:00', 1, '销售员A', '情人节订单'),
(1, 6, 1, 1, 35.00, 35.00, '2023-09-01', '2023-09-01 10:30:00', 1, '销售员A', '情人节订单'),
(2, 2, 2, 1, 88.00, 88.00, '2023-09-02', '2023-09-02 09:15:00', 2, '销售员B', '母亲节订单'),
(3, 9, 3, 2, 120.00, 240.00, '2023-09-03', '2023-09-03 14:20:00', 1, '销售员C', '金卡客户订单'),
(4, 5, 4, 1, 55.00, 55.00, '2023-09-04', '2023-09-04 11:10:00', 3, '销售员D', '开业花篮'),
```

```
(5, 1, 5, 1, 99.00, 99.00, '2023-09-05', '2023-09-05 15:45:00', 2, '销售员E', '生日订单'),
(5, 4, 5, 1, 77.00, 77.00, '2023-09-05', '2023-09-05 15:45:00', 2, '销售员E', '生日订单'),
(7, 10, 7, 1, 40.00, 40.00, '2023-09-07', '2023-09-07 16:30:00', 4, '销售员F', '探病订
单'),
(10, 9, 10, 1, 120.00, 120.00, '2023-09-10', '2023-09-10 16:20:00', 2, '销售员G', '钻石客
户订单'),
(10, 1, 10, 2, 99.00, 198.00, '2023-09-10', '2023-09-10 16:20:00', 2, '销售员G', '钻石客户
订单');
```

5.11 客户购买历史表 (flower_customer_purchase_history)

```
-- 客户购买历史表
DROP TABLE IF EXISTS flower_customer_purchase_history;
CREATE TABLE flower_customer_purchase_history (
    id BIGINT(20) NOT NULL AUTO_INCREMENT COMMENT '购买历史ID',
    customer_id BIGINT(20) NOT NULL COMMENT '客户ID',
    flower_id BIGINT(20) NOT NULL COMMENT '鲜花ID',
    order_id BIGINT(20) NOT NULL COMMENT '订单ID',
    quantity INT(11) NOT NULL COMMENT '购买数量',
    unit_price DECIMAL(10,2) NOT NULL COMMENT '单价',
    total_amount DECIMAL(10,2) NOT NULL COMMENT '总金额',
    purchase_date DATE NOT NULL COMMENT '购买日期',
    purchase_time DATETIME NOT NULL COMMENT '购买时间',
    category_id BIGINT(20) NOT NULL COMMENT '分类ID',
    preference_match TINYINT(1) DEFAULT 0 COMMENT '是否匹配客户偏好 1-匹配 0-不匹配',
    create_by VARCHAR(64) DEFAULT '' COMMENT '创建者',
    create_time DATETIME DEFAULT CURRENT_TIMESTAMP COMMENT '创建时间',
    update_by VARCHAR(64) DEFAULT '' COMMENT '更新者',
    update_time DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP COMMENT
'更新时间',
    remark VARCHAR(500) DEFAULT NULL COMMENT '备注',
    PRIMARY KEY (id),
    KEY idx_customer_id (customer_id),
    KEY idx_flower_id (flower_id),
    KEY idx_order_id (order_id),
    KEY idx_purchase_date (purchase_date),
    CONSTRAINT fk_purchase_customer FOREIGN KEY (customer_id) REFERENCES
flower_customer(id),
    CONSTRAINT fk_purchase_flower FOREIGN KEY (flower_id) REFERENCES flower_info(id),
    CONSTRAINT fk_purchase_order FOREIGN KEY (order_id) REFERENCES flower_order(id),
    CONSTRAINT fk_purchase_category FOREIGN KEY (category_id) REFERENCES
flower_category(id)
) ENGINE=InnoDB AUTO_INCREMENT=1 DEFAULT CHARSET=utf8mb4 COMMENT='客户购买历史表';

-- 插入10条模拟数据
INSERT INTO flower_customer_purchase_history (customer_id, flower_id, order_id,
quantity, unit_price, total_amount, purchase_date, purchase_time, category_id,
preference_match, remark) VALUES
(1, 1, 1, 2, 99.00, 198.00, '2023-09-01', '2023-09-01 10:30:00', 1, 1, '匹配客户偏好'),
(2, 2, 2, 1, 88.00, 88.00, '2023-09-02', '2023-09-02 09:15:00', 2, 1, '匹配客户偏好'),
(3, 9, 3, 2, 120.00, 240.00, '2023-09-03', '2023-09-03 14:20:00', 9, 1, '匹配客户偏好'),
(4, 5, 4, 1, 55.00, 55.00, '2023-09-04', '2023-09-04 11:10:00', 5, 1, '匹配客户偏好'),
(5, 1, 5, 1, 99.00, 99.00, '2023-09-05', '2023-09-05 15:45:00', 1, 1, '匹配客户偏好'),
(5, 4, 5, 1, 77.00, 77.00, '2023-09-05', '2023-09-05 15:45:00', 4, 0, '不匹配客户偏好'),
(7, 10, 7, 1, 40.00, 40.00, '2023-09-07', '2023-09-07 16:30:00', 10, 1, '匹配客户偏好'),
(10, 9, 10, 1, 120.00, 120.00, '2023-09-10', '2023-09-10 16:20:00', 9, 1, '匹配客户偏好'),
(10, 1, 10, 2, 99.00, 198.00, '2023-09-10', '2023-09-10 16:20:00', 1, 1, '匹配客户偏好'),
(6, 3, 6, 1, 66.00, 66.00, '2023-09-06', '2023-09-06 13:30:00', 3, 1, '匹配客户偏好');
```

6. 总结

本系统设计涵盖了鲜花销售的核心业务流程，包括商品管理、客户管理、订单管理、预定管理、库存管理和智能推荐等功能模块。通过合理的数据库表结构设计和初始化数据，为后续的系统开发奠定了坚实的基础。

系统充分利用了若依开源平台的优势，在此基础上扩展了智能销售功能，结合LangChain4j实现智能推荐，能够有效提升用户体验和销售转化率。新增的预定功能和完善的销售历史记录表，为系统提供了更全面的业务支持和数据分析基础。

所有业务表均以"flower_"作为前缀，与若依平台的系统表进行了明确区分，便于后续的维护和管理。同时，鲜花信息表中的图片字段采用JSON格式存储网络图片URL，支持多张图片的展示需求。为了便于移动端扫描和管理，我们在关键业务表（鲜花信息表、订单表、预定表）中添加了二维码存储字段，支持生成和存储对应的二维码信息。